1 Abstract

Classical *Schrödinger Bridge* (SB) problem seeks a path measure that minimally deviates from a given reference measure subject to prescribed initial and terminal marginals. This works investigate numerical approach to dynamic of interacting brownian particle with normal reflection term in low-dimensional setting ($d \leqslant 3$) on bounded domain $\mathcal{D} \subset \mathbb{R}^d$. To alleviate the difficulty due to nonlinearity incurred by non-local coupling, a nested fixed-point is adopted with Sinkhorn-type iterates. Further, fully semi-lagrangian discretization is used to propagate the solution of forward-backward partial differential equations (FB-PDEs).

2 Introduction

3 Related Works

Interest in spatially-constrained SB is largely under-developed, especially when also incorporating mean-field framework. This work is inspired by [**?** **?**]. First paper studies SB with reflected dynamic with Sinkhorn iterate where the time marching solution involve Neuman and Rudin boundary conditions, and show great performance in low-dimension. The latter generalized standard Sinkhorn to accept partial-integro-PDEs associated with mean-field SB resulting in nested version. Along this line [**?**] studies degenerate, sub-Riemannian dynamics, and use Sinkhorn simiarl to the former. Recent work [**?**] use Finite element discretization to solve the weak form of coupled PDEs for reflected SB. [**?**] derives forward-backward SDEs associated coupled FB-PDEs of reflected SB and extend divergence-based likelihood training in smooth bounded domain and shown to scale in high-dimension. Discrete-time framework also been explored with chance-constrants [**?** **?**] with mixture of Guassian marginal contraints. Alternative method such as introducing direct soft-penalty for particle escaping region exists as demonstrated in [**?**] using FB-SDE representation. There is a direct link between system of optimality of SB and MFG, therefore one can naturally adopt a scheme in MFG such as finite-difference to solve SB problem. [**?** **?** **?**] adopt a semi-lagrangian approach.

4 Formulation

5 Numerical Scheme